Transition Risk, in Plain Terms

What the tool measures, what's in scope, and how the numbers are made

A non-technical companion to the full BSR Transition Risk Methodology. Written for a reader who is new to climate transition risk. Screening-grade tool — not investment advice or an assured regulatory disclosure.

1. The one-paragraph version

As the world moves away from fossil fuels, companies face costs that have nothing to do with storms or floods — they come from the *transition itself*: carbon prices, cheaper clean competitors, more expensive supply chains, and shifting reputations. This tool takes a company (or a portfolio of assets) and estimates, year by year from 2025 to 2050 and under different climate-policy scenarios, how much money that transition could cost — and how much of the company's asset value could be stranded (made worthless early). It breaks the answer into **four clearly separated channels** so you can see *where* the risk comes from, not just a single black-box number.

2. What you put in

For each entity or asset you provide a short profile:

Input	Plain meaning
Sector	What the business does (e.g. coal power, steel, real estate). Picks the right physics and cost curves.
Emissions (Scope 1, 2, 3)	How much CO_2 it emits directly (1), from its electricity (2), and across its value chain (3). Scope 3 is split into two: **upstream** (suppliers → supply-chain cost, Layer 3) and **use-phase** (what customers emit using its products → product-demand risk, Layer 2).
Annual revenue	Used to scale supply-chain and reputation effects.
Transition plan (optional)	Planned pivot capex and a "how well-placed today" positioning score — how the company adapts, not just how exposed it is (see Layer 2).
Replacement value	What the physical assets are worth — the thing that can be "stranded."
Region	Which carbon-price band **and jurisdiction stringency** (EU ETS vs US vs Canada …) and supply-chain structure apply. Country or sub-national code.
Decarbonization target (optional)	A net-zero year, plus an optional **residual %** — how far short of full net-zero the target lands (e.g. 20% = an 80% cut), so a partial target still carries carbon on what's left.

Input	Plain meaning
Attribution % (optional)	For an investor/lender: what share of the asset is *yours*.

Then you pick one or more **scenarios** — coherent stories about how fast the world decarbonizes, from "Net Zero 2050" (fast, orderly) to "Current Policies" (slow). These are the standard **NGFS** scenarios central banks use.

When a company barely discloses. If you only know a company's *sector, country and revenue* — a private firm, an emerging-market producer — the tool can **estimate its Scope 1+2 emissions** from the sector's carbon intensity (tCO_2 per $ of output) × its revenue, and fill in everything else from model defaults (positioning, capex, supply-chain cost). So an opaque entity still gets a full screening picture. Estimated figures are flagged as such and are always overridden the moment you enter reported data.

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3. What comes out

- **A cost timeline** (2025–2050) for each scenario: the annual cash-flow hit from transition, split into the four channels.
- **A stranded-asset estimate**: how much replacement value is written off, and when.
- **A climate-adjusted valuation** (optional DCF): the company's value with vs without transition risk.
- **Uncertainty and sensitivity**: a range around the central number, and a ranking of *which assumption matters most*.
- **Alignment metrics**: financed emissions (PCAF), an Implied Temperature Rise, and how the portfolio tracks against a 1.5 °C pathway.
- **A decarbonization lever map**: which specific abatement options exist for each sector, and where the company's transition plan has gaps.

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4. The four channels (this is the core idea)

The tool follows the four transition-risk categories defined by the **TCFD** (the global climate-disclosure framework). Each is one "layer," each is computed separately, and — this is important — **each is routed once by design**, with a couple of disclosed boundary exceptions (see §5) rather than a blanket "counted exactly once" guarantee.

Layer 1 — Policy & Legal risk (the carbon-price channel)

Question it answers: *What does it cost to emit, once emissions carry a price?*

Simple math:

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carbon cost  =  emissions  ×  carbon price
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But a firm rarely eats the whole bill. Two adjustments: - **Pass-through** — a company with pricing power passes some cost to customers. That passed-on part isn't its loss (it becomes a supply-chain effect for others — see Layer 3). - **Free allowances** — some emitters get free permits. Combined with pass-through, a firm can occasionally come out *ahead* (a "windfall") — the math allows this rather than hiding it.

The carbon price itself rises over time and differs by scenario and region; we read it from the published **NGFS** price tables and interpolate between years.

Geography matters, and now it bites. The NGFS tables set price by broad band (advanced / emerging economies), which alone would charge a German mill, a US mill and a Japanese plant the *same* price. On top of the band we apply a **jurisdiction stringency factor** reflecting how real carbon policy differs today — the **EU ETS** and UK sit *above* the band, the **US and Canada** *below* — converging back to the band by ~2040 in ambitious scenarios (which assume policy harmonization) but staying divergent in the weak ones. So a German site now correctly carries a heavier near-term carbon cost than a Canadian one. (Enter a country, e.g. `DEU`, `CAN`, `USA`.)

Layer 2 — Technology risk (the "cheaper competitor" channel)

Question it answers: *When does the clean alternative get cheap enough to make my asset obsolete — and how much value do I lose when it does?*

Two ideas: 1. **Learning curves (Wright's Law).** Every time the world doubles how much solar (or batteries, or green hydrogen) it has built, the cost drops by a fixed percentage. So we can project when the *challenger* technology's cost falls below the *incumbent's*. That year is the **crossover** — the tipping point. **And it's location-specific:** a watt in Texas or the Gulf is far cheaper than in Northern Europe or Japan, so those sunny/windy regions reach crossover — and produce green steel, ammonia and cement — *years earlier*. The model applies a geographic resource/cost factor by region, with **sub-national resolution for the US and Europe** — where a watt and a transition plan really do look different in Texas vs California, or Spain vs Germany. Same steel plant, green-steel cost parity: US Southwest ~2026, Texas ~2027, California ~2034, US Northeast ~2039; Spain ~2031, Germany ~2040, Japan ~2042. (Enter a region as `USA-TX`, `USA-CA`, `ESP-S`, etc. for the finer view.)

Adaptability (this matters a lot). A company isn't frozen — under a transition scenario it *migrates* toward the low-carbon business. So instead of assuming an ICE carmaker simply loses all its revenue, the model lets it **pivot**: it captures part of the green upside (offsetting the loss), spends **transition capex** to get there, and — if it's already partly transitioned — has less to strand. How much it offsets depends on two things: its **ambition** (defaulted from the scenario — a Net-Zero world pulls companies into an aggressive pivot; a Current-Policies world doesn't), and its **present positioning** (how well-placed it is today, from its emissions, the maturity of its sector's decarbonization options, and the strength of its transition plan — with a manual override for the judgement calls). A well-positioned, ambitious carmaker's exposure roughly halves; a poorly-positioned laggard keeps most of the loss *and* pays more to catch up. This is "here's how the company looks if it follows this pathway, from where it starts today" — not "here's the damage if it does nothing." The capex is the honest price of that pivot, so a transition plan is never free.

1. **Stranding.** Once crossover hits (or demand for the old product simply collapses), the asset loses value along an **S-curve** — slow at first, then fast, then leveling off — because plants don't shut overnight. How fast depends on the sector (power flips quickly; heavy industry and buildings turn over slowly).

We also carry an *uncertainty band* on the crossover year (costs are forecasts, not certainties) and can optionally make the crossover happen sooner when the carbon price makes the dirty option more expensive.

Products that emit when customers use them. For a company that *makes* carbon-emitting products — diesel engines, construction machinery, petrol cars, fuels — the biggest risk isn't its own factories; it's that its *customers* face carbon costs and switch to cleaner alternatives, so demand for the dirty product line falls. If a company reports these "use-of-sold-products" emissions, we translate a slice of that future customer carbon burden into demand-and-margin pressure on the maker. For Caterpillar this is the single largest effect — correctly flagging that its diesel machinery line is what's exposed, even though its own plants are relatively clean.

Layer 3 — Market risk (the supply-chain channel)

Question it answers: *Even if I'm clean, how much do my inputs cost more because my suppliers are paying for carbon?*

We use a standard tool from economics — an **input-output table** (who buys from whom across 20 sectors) — and the **Leontief inverse**, which traces a cost increase in one sector through *all* the ripples it causes upstream. If steel gets more expensive, carmakers feel it; if power gets more expensive, everyone feels it. We add up the ripples that land on your sector. (Optionally the firm passes part of this cost on again, so only its net share counts.) A higher-resolution version splits this across 49 world regions using the **EXIOBASE** global trade database.

If a company reports its own supply-chain emissions (upstream Scope 3), we price *that* disclosed number directly instead of estimating it — so BASF's 90-million-tonne value chain or Nike's 9.5 million tonnes drives the cost, not a generic sector average. The estimated version is the fallback for firms that don't report.

Biogenic inputs aren't fossil. For forest-products firms (pulp, paper, timber), much of the upstream footprint is *biogenic* — carbon in sustainably-managed wood fibre that the forest re-absorbs — not fossil carbon. We net that share out before pricing, so a pulp mill's wood supply isn't charged as if it were steel or cement.

Layer 4 — Reputation risk (the cost-of-capital channel)

Question it answers: *Do investors and lenders charge me more because of my climate exposure?*

Academic research (**Sautner et al., 2023**) measured how a company's "climate change exposure" is priced into its cost of capital. We map each sector to that measure and translate it into an adjustment to the discount rate. Crucially, the sign cuts both ways: a company with high **downside** exposure (regulatory + physical risk, e.g. a coal utility) pays *more* for capital (≈ +27 bps), while a climate **winner** with high opportunity exposure (e.g. a renewables developer) is financed *more cheaply* (≈ −33 bps) — consistent with the "green premium" evidence that low-carbon firms enjoy a lower cost of capital. By default we route this to the cost of capital, because that's the part the research supports; the revenue route is a clearly-labeled *manual* overlay.

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5. The golden rule: route each risk once (with two disclosed exceptions)

The biggest way climate models mislead is by **double-counting** — charging the same dollar twice. The architecture routes each channel to one destination, with two honestly-flagged boundary cases rather than a blanket guarantee:

Channel	Where it lands (once)
Layer 1 — carbon	Operating cost
Layer 2 — technology	Revenue erosion **+** a *separate* asset write-down
Layer 3 — supply chain	Input cost
Layer 4 — reputation	Cost of capital **or** revenue — never both

For example, the carbon a firm passes on to customers is removed from *its* cost (Layer 1) and only enters the supply chain (Layer 3) — never both. Stranded value (a balance-sheet write-down) is kept in its own column and never added into the cash-flow cost total. And a company's **purchased-electricity carbon (Scope 2)** is charged only once — through the higher power prices in Layer 3, not also as a direct Layer-1 bill (unless it actually pays an explicit carbon charge on that electricity).

The two disclosed exceptions. (1) A demand collapse shows up as *both* eroded revenue (cash flow) and a stranded write-down (balance sheet) — these are two **lenses on the same loss**, shown side by side and never summed. (2) The "% stranded" figure is defined as the write-down actually recognised by 2050, so it always matches the dollar impairment; a separate "strandable ceiling" shows the theoretical maximum.

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6. Turning it into a value, and putting a range on it

- **Discounting.** A cost in 2045 hurts less than the same cost today, so future costs are discounted back to present value — using a *real* (inflation-adjusted) rate so we don't mix nominal cash flows with real ones.
- **Uncertainty (Monte-Carlo).** We re-run the model hundreds of times, each time nudging the carbon price, pass-through, and supply-chain elasticity, to get a P5–P95 range. This range is deliberately labeled a **conditional floor**: it's the spread *within* a chosen scenario, not the full uncertainty (which also includes *which* scenario comes true).
- **Sensitivity (tornado).** We swing each assumption one at a time to see which one moves the answer most — so you know which input is worth improving. There are two views: one for the cash-flow cost, one for the stranding (which is driven by different assumptions: trigger year, S-curve speed).

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7. Beyond the number: the decision layer

The tool doesn't stop at "here's the cost." It also helps answer *what to do*:

- **Financed emissions & a temperature score** — *screening indicators* that approximate the shape of PCAF financed emissions and an Implied Temperature Rise (1.5 °C benchmark). They are **not** the certified PCAF / SBTi / PACTA metrics and shouldn't be reported as compliant with them.
- **Marginal Abatement Cost Curves + optimizer** — given a budget, which emission cuts buy the most reduction per dollar (cheapest-first).
- **Decarbonization Lever Library** — for each sector, the concrete options to cut emissions (renewables, heat pumps, green hydrogen, CCUS, regenerative agriculture…), positioned by where they sit in the value chain (your suppliers / your operations / your customers), with a "just transition" note on nature-and-people impacts. Overlay a company's actual plan to see which core levers are missing. This is a **structured reference**, not a score.
- **Lever-level capital plan** — a bottom-up build of the transition capex. For each lever the company is exposed to, the tool shows its **current adoption**, a **target**, the **abatement** it can address, and the **forward capex** to close the gap. Those per-lever costs *add up* to the transition capex instead of a single top-down number. It's seeded from the model's estimate and then **fully editable** — adoption, cost, and the year-by-year **capital-planning** phasing — and the edited build flows straight into the cost model when you apply it. This is how "what would it cost us to actually do this" becomes a first-class, editable input rather than an assumption. The seed puts capital where it really goes: it weights each lever by how **capital-heavy** it is per tonne (a hydrogen-DRI or CCUS plant, not cheap efficiency), so a steelmaker's plan leads with the DRI switch and an automaker's with fleet electrification — matching how those companies' published plans actually allocate money.

- **Opportunity lens** — for a company in a sector that *grows because of* decarbonization (renewables, EV, transition minerals, mass timber — where Net-Zero demand rises well above baseline policy), the tool flags it as a **transition beneficiary** and puts a separate, screening-grade number on the **growth capital** to expand that low-carbon business. This is an *opportunity*, shown alongside the risk and **never added into** the transition-risk total — so a clean-growth company reads as the winner it is, not just a cost.
- **Disclosure export** — a report structured against **IFRS S2** and **ESRS E1**, the two main climate-disclosure standards. (The experimental supply-chain "contagion" amplifier is deliberately excluded from this export.)

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8. Where the data comes from

Piece	Source
Carbon prices by scenario/region/year	**NGFS Phase V** (the central-bank scenario set)
Technology cost/learning rates	**IRENA 2024**, **Lazard 2025 v18**, **BNEF 2024**; forecast method from **Farmer & Lafond** (2016) and **Way, Ives, Mealy & Farmer** (2022, Oxford)
Supply-chain structure	**EXIOBASE-3** global input-output database
Reputation → cost of capital	**Sautner et al. 2023** (*Journal of Finance*)
Sector abatement options & costs	**IPCC AR6**, **IEA**, Mission Possible Partnership

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9. What this is — and honestly isn't

It is: a transparent, scenario-based *screening* tool that separates the four transition channels, avoids double-counting, shows its working, and ranges its answers.

It isn't: - A precise firm-level forecast. Many inputs are sector medians (pass-through, reputation exposure), not company-specific — the tool flags where firm data would sharpen the answer. - A physical-climate model. Floods and heat are a *separate* tool; this one is transition only. - A tail-risk / Value-at-Risk engine. The uncertainty range is a conditional floor, and we label it as such rather than dressing it up as full VaR. - A finished regulatory disclosure. It structures the output against IFRS S2 / ESRS E1, but a real filing needs specialist review and assured data. - A full real-estate transition model. Buildings show near-zero here because only carbon pricing is modelled; performance standards, retrofit capex, and green-vs-brown rent effects are not yet built — so a low real-estate number is *not* a clean bill of health. - A free-decarbonization model. Setting a net-zero target lowers future carbon cost, but the **capital and operating cost of the abatement** to get there is analysed separately (in the abatement/MACC view) and is not automatically netted into the headline — treat a target as reducing gross exposure, not as costless. - A technology-equivalence oracle. Layer 2 compares a challenger's cost to an incumbent's, but those costs aren't always like-for-like (e.g. solar power vs coal power ignores firming/grid value) — the crossover year is indicative, not a dispatch-accurate parity date. - A group-consolidation model. A diversified firm (multiple business lines) is rolled up as the **sum** of its lines — each line correctly transitions on its own sector/geography, but group-level correlation, cross-subsidy and a single optimised group plan are not modelled. - A full growth-capex model. The transition *risk* capex is the cost to **decarbonize the company's own emissions** — not the full growth capital a clean-energy company spends to expand its low-carbon business. A renewables/grid utility (e.g. Iberdrola) therefore shows a small decarb-pivot capex even though it runs a very large capital programme, and enabling infrastructure (grid) with no direct abatement of the firm's own footprint sits outside the lever capital plan. The **opportunity lens** (see §7) puts a *separate*, screening-grade number on that growth capital, but it is not a bottom-up capital budget.

The tool covers **30 sectors** — the 20 core plus seven from the BSR lever library (apparel, consumer goods, financial services, healthcare, professional services, telecom, and metals & mining), industrial & construction equipment (a diesel-machinery maker whose risk is the use-phase of its products), and two forest-products sectors — **pulp & paper** (a biomass-powered low-carbon producer) and **solid wood / mass timber** (a transition **winner** — mass timber displaces steel and cement, so demand grows). Metals & mining and mass timber are treated as transition **winners** (demand grows, cheaper capital), not stranding risks. Supply-chain (Layer 3) cost is scaled to a firm's actual **bought-in inputs**, so an asset-light, high-revenue business (a bank) no longer shows an implausible supply-chain carbon bill.

The guiding principle throughout: **be useful and be honest about the limits.** Every place where a figure is a proxy, an assumption, or an opt-in refinement is labeled as such — in the tool, the audit trail, and the full methodology document. Every engine result also carries a **data-quality flag** (firm / sector-proxy / degraded) so a number built on placeholders is never mistaken for one built on company data.